

Project Map

1. Image Capture

2. **Preprocessing** (Noise reduction, white balance correction, ROI selection, scale calibration)

3. Segmentation

Acne: Threshold + watershed → lesion detection.

Pigmentation: Color threshold in LAB/HSV → pigmented mask.

Wrinkles: Edge detection + skeletonization → wrinkle lines.

Erythema/sensitivity: Red channel thresholding.

4. Parameter Extraction

Measurable Parameters

Measurable Parameters

Acne

Parameter	Goal	Output	Quantitation
Lesion count (no/cm2)	How many lesions are present per unit area	Count expressed as absolute numbers	Mild (<20 comedones, <15 inflammatory), Moderate (20–100 comedones, 15–50 inflammatory), Severe

			(>100 comedones, >50 inflammatory)
Lesion type classification	Lesions categorization	non-inflammatory lesions: comedones (whiteheads, blackheads). Inflammatory lesions: papules, pustules, nodules.	Pustule — area 0.3–5 mm², circularity > 0.6, mean center intensity high (bright spot). Papule — area 0.2–3 mm², circularity 0.4–0.8, no bright center. Nodule — area > 5 mm² and/or low circularity (<0.4). % Inflammatory = (count_pustules+count_nodules)/N ×100. Suggested bands: Low inflammatory % <20%; Moderate 20–50%; High >50%.
Lesion density	Measure local clustering	Count lesions in a defined Region of Interest (ROI), e.g., cheek, forehead. <i>Number of lesions / cm²</i>	Mild (<1 lesion/cm²), Moderate (1–3 lesions/cm²), Severe (>3 lesions/cm²).
Redness index (Erythema intensity) (Simple color analysis using a standard camera)	Quantify inflammatory redness objectively and normalize for lighting/skin tone.	erythema index /erythema severity	Either Method A or Method B depending on feasibility Scale: < 15% = mild

			15-35% = moderate > 35% = severe
Pore size & density	Count visible pores in a defined Region of Interest (ROI), e.g., cheek, nose, forehead.	Density = Number of pores / cm ² (or per standardized ROI image). Size = Average pore diameter (µm) in ROI.	Mild: Density <50 pores/cm ² , Size <150 µm, Moderate: Density 50–100 pores/cm ² , Size 150–250 µm, Severe: Density >100 pores/cm ² , Size >250 µm

EI quantifies the degree of redness in skin compared to normal baseline. Redness = proxy for inflammation / erythema in acne. Computed from RGB image intensities.

We can use either method:

Method A:

$$\% \text{ NR} = (\text{R}_{\text{lesion}} - \text{R}_{\text{bg}}) / \text{R}_{\text{bg}}$$

R_{lesion} = mean red channel intensity of lesion pixels.

R_{bg} = mean red channel intensity of adjacent unaffected skin.

Outputs percentage redness increase relative to baseline skin.

Method B:

Standard Erythema Index

$$EI = (R-G)/(R+G+B)$$

Normalizes redness against brightness (R+G+B).

Subtracts green channel to cancel non-red effects.

5. Severity Classification

Pigmentation

Parameter	Goal	Output	Quantitation
Pigmentation area %	Percentage of the skin region affected by pigmentation (brown/dark lesions) in a defined Region of Interest (ROI), e.g., cheek, forehead.	% pigmentation = number of pigmented pixel (lesion area)/ total number of pixels in ROI X 100 ROI e.g. cheek, forehead, etc	Mild: $\leq 10\%$ Moderate: 10 - 30 % Severe: $\geq 30\%$
Pigmentation intensity/ Δ Gray Value	Assesses pigmentation darkness compared to adjacent normal skin	Δ Gray = mean intensity (pigmented ROI) - mean intensity (normal skin ROI)	Mild contrast Δ Gray = 0-20 Moderate contrast Δ Gray = 20-50 Severe contrast Δ Gray > 50
Skin Hyperpigmentation Index (SHI) which is equivalent to MASI	Skin Hyperpigmentation Index (SHI) instead of MASI.	https://drive.google.com/drive/folders/1TVvquhgauYLC-31FgaAVXhg3TiVVM-sb https://shi.skinimageanalysis.com/ As discussed	1: no hyperpigmentation 1-2: light hyperpigmentation 2-3: Medium hyperpigmentation 3-4: Severe hyperpigmentation 4: Maximum hyperpigmentation

Wrinkle

Parameter	Goal	Output	Quantitation
Wrinkle Count (no./cm²)	Quantify how many wrinkles are present in given ROI	Wrinkle Count (no./cm ²) = no. of detected wrinkle lines in ROI/ ROI in cm ² *Edge detection / line segmentation (python canny/hough)	Mild: < 5 /cm ² Moderate: 5-10/cm ² Severe: > 10/cm ²
Wrinkle length (mm or pixel/cm)	Measure the length of each wrinkle line	Use skeletonized wrinkle lines → measure lengths → convert pixels → mm using scale	Mild : < 20 mm average length Moderate: 20 - 40 mm Severe: > 40 mm
Wrinkle depth/ Δ Gray	Measure wrinkle depth by intensity contrast b/n wrinkle trough and surrounding skin	Select wrinkle ROI + adjacent ROI → compute mean gray values → subtract. Larger Δ Gray = deeper wrinkle.	Mild: < 20 Moderate: 20-50 Severe: > 50
Wrinkle Density %	Proportion of ROI occupied by wrinkle pixels.	Binary segmentation => wrinkle pixel / ROI pixel X 100	Mild: < 5 % Moderate: 5 - 15 % Severe: > 15 %

Acne:**Mild:**

Acne count < 6/cm², mostly comedones, few papules

Redness Index: < 15%

Mostly non-inflammatory and small clusters

Moderate:

Acne count: 6-15/cm², Mix of papules & pustules

Redness Index: 15–35%

Visible Inflammation

Severe

Acne count: >15/cm², Nodules, cysts, burst/pus-filled lesions

Redness Index: >35%

Mostly inflammatory

Pigmentation**Mild:**

% Pigmented Area: <10%

Δ Gray (intensity difference): <20

Melanin Index (MI): < 50

Small patches, light scale

Moderate:

% Pigmented Area: 10-30%

Δ Gray (intensity difference): 20-50

Melanin Index (MI): >50

Wider patches, dark tone

Severe

% Pigmented Area: >30%

Δ Gray (intensity difference): >50

Melanin Index (MI): >100

Extensive, high contrast pigmentation

Wrinkles

Mild

Wrinkle Count (no./cm²): <5/cm²

Length (mm): < 20 mm

Depth (ΔGray intensity): < 20

Fine Line, shallow

Moderate:

Wrinkle Count (no./cm²): 5-10/cm²

Length (mm): 20-40 mm

Depth (ΔGray intensity): > 40

More defined lines, moderate depth

Severe

Wrinkle Count (no./cm²): >10/cm²

Length (mm): > 40 mm

Depth (ΔGray intensity): > 40

6. Severity Classification

Condition	Mild	Moderate	Severe
Acne (Lesion count, lesion type, redness index)	Lesion count: <6/cm ² Type: Mostly comedones, few papules Redness index (RI): <15%	Lesion count: 6–15/cm ² Type: Mix of papules & pustules RI: 15–35%	Lesion count: >15/cm ² Type: Nodules, cysts, burst/pus-filled lesions RI: >35%

Pigmentation (Pigmented area %, Δ Gray, Melanin Index)	Pigmented area: <10% Δ Gray: <20 MI: <50	Pigmented area: 10–30% Δ Gray: 20–50 MI: 50–100	Pigmented area: >30% Δ Gray: >50 MI: >100
Wrinkles (Wrinkle count, length, depth)	Wrinkle count: <5/cm ² Length: <20 mm Depth (Δ Gray): <20	Wrinkle count: 5–10/cm ² Length: 20–40 mm Depth: 20–50	Wrinkle count: >10/cm ² Length: >40 mm Depth: >50

7. Decision and Remedies

Acne

Parameter(s)	Severity Band	Recommendation Type	Examples (Home remedy / Cosmeceutical)	Duration
Lesion count (no./cm ²) + Redness Index	Mild (<6/cm ² , RI <15%)	Home remedy	cleanser, tea-tree oil, honey mask, diet advice (e.g. low sugar, etc)	2–4 weeks
	Moderate (6–15/cm ² , RI 15–35%)	Marketed OTC cosmeceutical	Benzoyl peroxide 2.5–5% gel, Niacinamide serum, Salicylic acid 2% cleanser	4–8 weeks
	Severe (>15/cm ² , RI >35%)	Referral alert	Advise dermatologist consultation	—

Pigmentation

Parameter(s)	Severity Band	Recommendation Type	Examples (Home remedy / Cosmeceutical)	Duration
% Pigmentation area, Melanin Index, Δ Gray	Mild (<10% area, MI <50)	Home remedy	Aloe vera gel, green tea extract	2–4 weeks

	Moderate (10–30% area, MI 50–100)	Marketed OTC cosmeceutical	Vitamin C serum, Niacinamide 5–10%	4–8 weeks
	Severe (>30% area, MI >100)	Referral alert	Advise dermatologist consultation	—

Wrinkles

Parameter(s)	Severity Band	Recommendation Type	Examples (Home remedy / Cosmeceutical)	Duration
Wrinkle count (no./cm ²), Avg. length, Depth Δ Gray	Mild (<5/cm ² , shallow)	Home remedy	Hydration (aloe vera, honey), cucumber mask, sunscreen use	2–4 weeks
	Moderate (5–10/cm ² , moderate depth)	Marketed OTC cosmeceutical	Retinol 0.1–0.3%, Hyaluronic acid serum	4–8 weeks
	Severe (>10/cm ² , deep ΔGray >50)	Referral alert	Advise dermatologist consultation	---

□ How matrix is going to work in prototype

- Input: Device calculates severity band from measured parameters
- Lookup: Software check severity band in above decision matrix
- Output: Display recommended home remedy/ OTC cosmeceutical + treatment duration
- Progress tracking: Compare before vs After parameters over duration
- Disclaimer: “For Information Use only, consult dermatologist ”

8. Decision matrix

 **experimental_matrix.xlsx**

9. Output

a. Severity

b. Remedies and duration

c. Duration

10. After completion of duration, before and after image comparison